LFM Pulse Compression for a 200 km Radar Target

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RESULT IN BRIEF. At fixed peak power, the 100 µs LFM pulse delivers +10.18 dB measured matched-filter SNR gain over the 10 µs pulse, while both retain 37.5 m nominal range resolution. A 200-trial production-chain study finds +9.99 dB mean gain (95% confidence interval for the mean: 9.88–10.10 dB).

I. PROBLEM STATEMENT & MATHEMATICAL FORMULATION

A. Radar Energy vs. Resolution Conflict: The Track 1 synthetic target has range R = 200 km and RCS $\sigma = 1.0\ m^2$. Detecting its weak echo requires sufficient radiated pulse energy $E_t = P_t \cdot \tau_p$. For unmodulated rectangular pulses, range resolution is constrained by pulse duration: $\Delta R_{unmod} = c \cdot \tau_p / 2$. Widening an unmodulated pulse from 10 µs to 100 µs scales energy by +10 dB but degrades nominal range resolution from 1.5 km to 15.0 km. Applying Linear Frequency Modulation (LFM) decouples resolution from pulse width:

$$\Delta R_{\text{unmod}} = \frac{c\tau_p}{2} \quad \longrightarrow \quad \Delta R = \frac{c}{2B} = \frac{3.0 \times 10^8\ \text{m/s}}{2 \times 4.0 \times 10^6\ \text{Hz}} = \mathbf{37.50}\ \text{m} \qquad (1)$$

B. LFM Chirp & Matched Filter Pulse Compression: A linear chirp sweeps frequency over sweep bandwidth B = 4.0 MHz within duration τ_p with chirp slope $k_r = B / \tau_p$. The receiver matched filter h(t) = s*(-t) compresses the echo into a narrow mainlobe. Its nominal range resolution is set primarily by bandwidth B:

$$s(t) = \text{rect}\left(\frac{t}{\tau_p}\right) e^{j\pi k_r t^2},\ k_r = \frac{B}{\tau_p} \quad \Longrightarrow \quad y(t) = s(t) * s^*(-t) \approx \tau_p \cdot \text{sinc}(\pi B t) \qquad (2)$$

C. Illustrative S-Band Link Budget: We assume P_t = 250 kW, G = 40 dBi per antenna, λ = 0.1 m [3.0 GHz], L = 3.0 dB, and F = 2.5 dB. These are simulation inputs, not published facility specifications. At R = 200 km, the Radar Range Equation yields:

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R_0^4 L} = \frac{(250\ \text{kW})(10^4)^2 (0.1\ \text{m})^2 (1.0\ \text{m}^2)}{(4\pi)^3 (200\ \text{km})^4 \cdot 10^{0.3}} = 3.946 \times 10^{-14}\ \text{W}\ (-134.04\ \text{dBW}) \qquad (3)$$

Receiver noise spectral density is $N_0 = k_B T_0 F$ = -201.48 dBW/Hz (7.120×10^{-21} W/Hz). Matched filter peak output SNR integrates total received energy $E_r = P_r \cdot \tau_p$. Comparing τ_2 = 100 µs against τ_1 = 10 µs proves the fundamental +10 dB SNR scaling law:

$$\text{SNR}_{\text{MF}} = \frac{P_r \tau_p}{N_0} \quad \Longrightarrow \quad \Delta\text{SNR}_{\text{MF}} = 10\log_{10}\left(\frac{\tau_2}{\tau_1}\right) = 10\log_{10}\left(\frac{100\ \mu s}{10\ \mu s}\right) = \mathbf{+10.00}\ \text{dB} \qquad (4)$$

II. PROPOSED METHODOLOGY & RADAR PROCESSING PIPELINE

The simulation executes a discrete-time processing chain: (1) Dual LFM chirp generation (B = 4 MHz, τ_1 = 10 µs, τ_2 = 100 µs) at f_s = 16 MHz; (2) Pulse train synthesis at PRT = 2.0 ms (PRF = 500 Hz, R_{unamb} = 300 km); (3) Target propagation channel at R = 200 km (round-trip delay τ_d = 1.333 ms) with complex circular AWGN; (4) Zero-padded FFT cross-correlation matched filtering preventing circular wraparound; (5) 1-D CA-CFAR detection (24 reference and 8 guard cells on each side, $P_{fa} = 10^{-6}$). The same peak power, target, and channel assumptions are used in both runs.

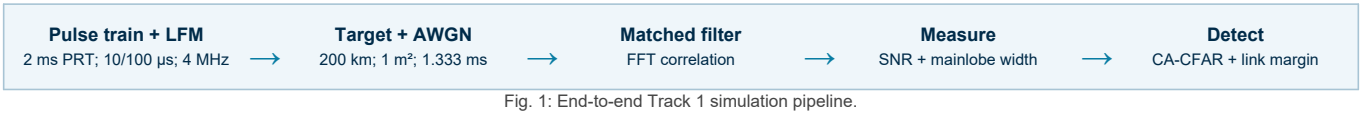


Fig. 1: End-to-end Track 1 simulation pipeline.

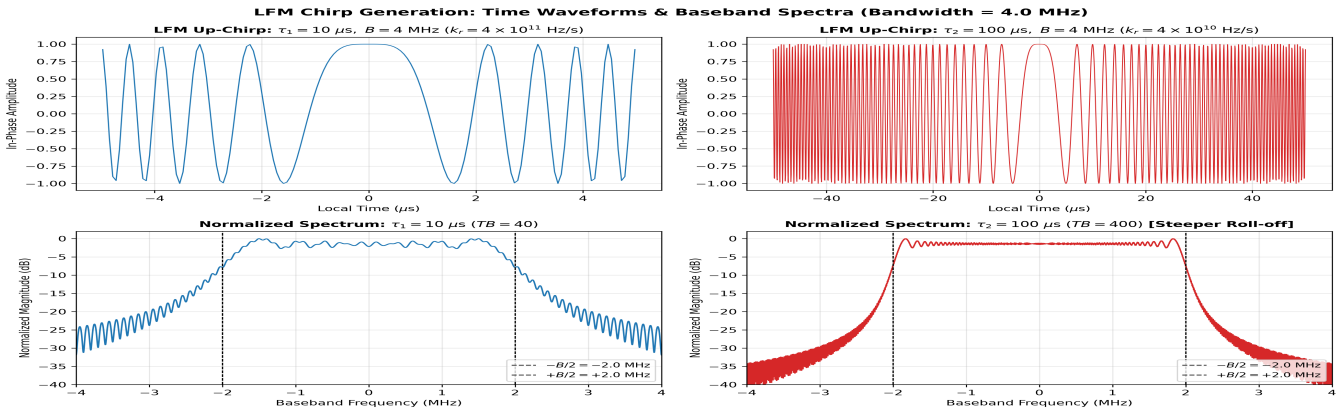


Fig. 2: Synthesized LFM chirp time-domain waveforms and normalized frequency spectra (B = 4.0 MHz) for 10 µs and 100 µs pulses.

III. QUANTITATIVE RESULTS, BENCHMARK COMPARISON & VALIDATION

Radar System Metric / Parameter	Pulse 1 (τ_1 = 10 µs)	Pulse 2 (τ_2 = 100 µs)	Enhancement / Comparison
Radiated Pulse Energy (P_t = 250 kW)	2.50 J	25.00 J	+10.0 dB
Theoretical Matched-Filter SNR	17.44 dB	27.44 dB	+10.00 dB predicted
Measured Peak SNR	17.13 dB	27.31 dB	+10.18 dB observed
Nominal Range Resolution c / (2B)	37.50 m	37.50 m	Same 4 MHz bandwidth
Measured Half-Power (-3 dB) Width	35.36 m	34.17 m	Similar compressed peaks
Theoretical Link Margin	+4.25 dB	+14.25 dB	Illustrative link model
CA-CFAR Target Detection Margin	+4.14 dB	+10.08 dB	Adaptive threshold

Validation: The fixed-seed core run measures +10.18 dB versus the +10.00 dB prediction. A separate 200-trial production-chain experiment measures +9.99 dB mean gain (95% confidence interval for the mean 9.88–10.10 dB). The small single-run offset reflects finite noise and sampling effects.

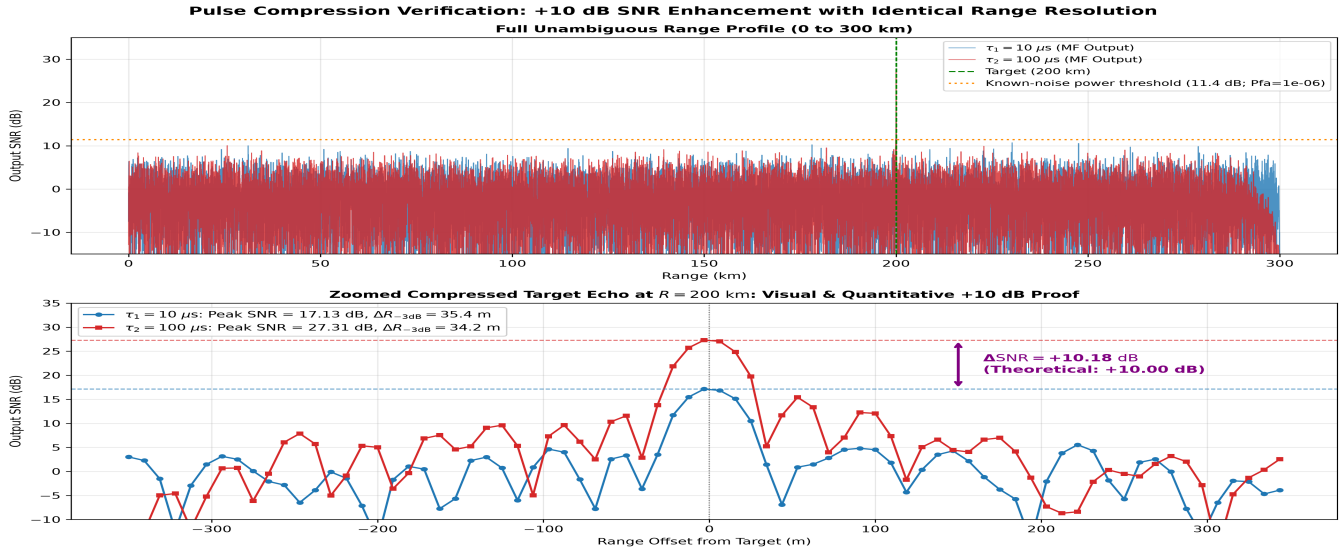


Fig. 3: Matched filter output at R = 200 km demonstrating the +10.18 dB SNR boost and similar 35 m mainlobe widths.

IV. KEY TAKEAWAYS, PRACTICAL CONSIDERATIONS & LIMITATIONS

1. Interpretation: At equal peak power, the 100 µs pulse carries 10× the energy. Matched filtering converts that energy into about +10 dB output SNR without broadening the compressed peak. The link and CFAR margins increase under the chosen illustrative parameters.
2. Minimum Range Blind Zone: Monostatic transmission blinds reception over $R_{min} = c\,\tau_p / 2$. For τ = 100 µs, R_{min} = 15.0 km (versus 1.5 km for 10 µs). A mixed short/long-pulse schedule is one possible design response.
3. Doppler Shift Coupling: Target radial velocity v_r induces Doppler frequency $f_d = 2v_r / \lambda$, causing an apparent range bias $\Delta R = -c\, f_d / (2\, k_r)$. Transmitting alternating up- and down-chirps can reduce this bias and estimate target velocity.
4. Scope: This is an AWGN point-target simulation, not a field measurement. Clutter, propagation, fluctuating RCS, and receiver imperfections can change detection performance. Reproduce the core plots with `python track1_lfm_simulation.py` and the 200-trial check included in the saved simulation results. The nominal CA-CFAR false-alarm model assumes independent cells, while oversampled filter outputs are correlated.